PROJECT FIELD GUIDE

Chinese Learning Game

What the folders and files do, and where to look when you want to change a feature. Line references match the project inspected on October 23, 2026.

```
ChineseGame/
|-- index.html                 shared pages and navigation shell
|-- styles.css                 main website styles
|-- world.css                  Social World layout and controls
|-- character.css              Character Studio styling
|-- package.json               dependencies and commands
|-- vite.config.js              Vite settings
|-- public/assets/models/       public FBX models and textures
|-- scripts/                    helper scripts
`-- src/
    |-- app.js                  navigation and main interactions
    |-- hsk1.js                 HSK 1 course and quizzes
    |-- profile.js              shared profile, rewards, local save
    |-- questions.js            original picture quiz content
    |-- world.js                 Three.js world and player controls
    |-- hsk.css                  HSK course styling
    |-- data/                    vocabulary, grammar, character catalog
    |-- utils/characterLoader.js model loading and appearance/animation
    |-- components/Character/    3D character preview
    `-- assets/models/           imported character models and textures
```

The big picture

The website shell lives in **index.html**. The application code in **src/app.js** switches views and loads the Study Mode or Social World. Both modes read and update **src/profile.js**, so the same device-local profile carries the player name, appearance, XP, coins, and course progress.

Architecture note: this project currently uses plain JavaScript with Three.js and Vite. It is not a React / React Three Fiber project at this time.

Start here

To change a visible page, find its section in index.html. To change behavior, find the matching JavaScript module. To change text or lesson material, edit the JSON/data file rather than the renderer.

1. Website shell and app behavior

index.html

Lines	What this section does
1-22	HTML document setup, fonts, stylesheets, and site header.
23-34	Main navigation buttons.
35-41	Main page content container and top bar.
42-53	Home view.
54-59	Study Mode containers: the HSK course and original picture practice.
60-66	Social World canvas, controls, interaction dialog, customization dialog, and profile dialog.
67-72	Character Studio and 3D preview/customization controls.
73-74	Loads src/app.js as an ES module.

src/app.js

Lines	What this section does
1-8	Imports profile rewards, picture questions, character data, and HSK styling.
10-22	Finds the page views and initializes app state.
24-63	Navigation between Home, Study, World, and Character Studio. Starts lazy-loaded course/world views.
65-85	Chooses and loads picture-practice questions.
87-108	Safe text escaping and question rendering.
110-141	Answer, feedback, retry/next actions, and rewards.
143-152	Navigation click handling.
154-173	Basic name and color customization dialog.
175-187	3D player profile preview.
189-end	Character Studio: selection, appearance changes, preview, reset, and save.

Small note: app.js imports hsk.css twice at lines 4-5. That import is redundant; it is not a second course or a second website.

2. Study Mode and shared player progress

src/hsk1.js

Lines	What this section does
1-9	Loads vocabulary and grammar data and defines lesson size, pass score, and initial course progress.
11-20	Mounts the HSK course and saves progress into the shared profile.
21-34	Escapes content, selects lesson words, calculates progress, uses browser Mandarin speech, and saves/renders.
36-57	Course home/dashboard and progress display.
59-62	Vocabulary lesson cards.
64-66	Grammar review screen.
68-74	Creates vocabulary and grammar quiz questions.
76-87	Starts and renders quizzes.
89-108	Scores quiz results and checks passing requirements.
110-115	Displays certificate information after passing the final assessment.
116-160	Chooses the active course screen and handles buttons.

Course data

Lines	What this section does
hsk1-vocabulary.json	2,410 lines. Word list with Hanzi, pinyin, meanings, and example sentences.
hsk1-grammar.json	45 grammar points with patterns, explanations, and example sentences.
src/questions.js	Lines 1-11 contain the original eight picture-practice questions. Each entry has category, picture/emoji, hint, answer, meaning, and choices.

src/profile.js

Lines	What this section does
1-2	Local-storage key and default profile/appearance.
4-21	Loads saved values, fills missing defaults, and migrates an older appearance.
23-25	Calculates level and title from XP.
27-32	Saves to browser storage and dispatches a profile-change event.
34-50	Updates visible profile labels, totals, name, and achievement text.
52	Returns profile data plus derived level/title.

Lines	What this section does
54-60	Awards Study Mode XP, coins, lesson count, and first-answer achievement.
62-70	Awards one-time Social World visit XP/coins.
72-77	Saves general profile updates and initializes visible profile values.

The profile is saved only in this browser/device. Account login and cloud synchronization are not configured.

3. Social World and character system

src/world.js

Lines	What this section does
1-5	Imports Three.js, profile rewards, character loading, and starts world creation.
6-24	Scene, camera, renderer, lighting, shadows, and color settings.
25-36	Shared materials and basic geometry.
38-53	Reusable helpers for blocks, spheres, and trees.
54-73	Building helper: walls, windows, door, roof, and sign.
75-102	Builds roads, plaza, classroom, other buildings, pond, pavilion, lanterns, trees, and flowers.
104-110	Avatar loading and failure status.
111-116	Reads saved avatar appearance, loads character, and places it in world.
118-131	Interactive destination data and glowing markers.
133-138	Tracks keyboard/touch movement, camera angle, jumping, interactions, and run state.
140-156	Shows location details; classroom navigates to Study Mode; other destinations can reward XP/coins.
158-230	Nearby location checks, keyboard/camera input, resizing, and touch buttons.
231-265	Animation loop: walking/running/jumping, animation state, and camera follow.
267-end	World cleanup when the view closes.

src/utils/characterLoader.js

Lines	What this section does
1-13	Imports loaders, skeleton cloning, catalog, model URLs, and textures.
14-29	Shared loaders, cache, catalog lookup, model URL map, and toon data.
30-98	Validates model choice and loads/applies Mage and Maid textures.
99-113	Loads models and caches them.
114-168	Prepares cloned materials and texture mappings.
170-185	Defines outfits and hair, skin, and eye options.
187-232	Applies appearance choices to model materials.
234-295	Creates character, loads model, handles rig setup, and reports status.
296-315	Changes animation state and blends available actions.
316-325	Changes outfit and appearance.
326-395	Updates mixer and procedural poses.
396-404	Disposes resources and exports catalog/customization choices.

src/components/Character/characterPreview.js

Lines	What this section does
1-5	Imports and preview entry point.
6-24	Preview scene, camera, renderer, lights, and display platform.
25-41	Loads or changes the preview avatar.
42-48	Fits preview rendering to its panel.
49-57	Preview animation/render loop.
58-63	Drag-to-rotate input.
64-end	Stops animation and releases resources.

4. Data, assets, setup, and feature boundaries

Character catalog and model files

Lines	What this section does
1-8	Catalog schema and four free character records: their IDs, presentation, model paths, outfits, and palettes.
9-10	Premium cosmetics are listed but disabled pending store/entitlement setup.
12-18	Supported animation labels and customization options.
19-20	End of catalog.

Active character-to-model mappings are near lines 21-23 of characterLoader.js. The imported model files and textures live under **src/assets/models/characters/**; FBX files and their textures also live under **public/assets/models/characters/**. Older versioned GLBs are still in the folder, but the mapping tells you which assets the loader actually uses.

Styling and project configuration

Lines	What this section does
styles.css	Main site styling. It is compressed into one long physical line, so section-by-section line references are not useful.
world.css	Social World canvas/layout and responsive world controls.
character.css	Character Studio and preview styling.
src/hsk.css	HSK course, lessons, quizzes, and certificate styling.
package.json, lines 1-17	Project metadata, npm scripts, Three.js dependency, and Vite development dependency.
vite.config.js, lines 1-20	Vite dev server/watch behavior and production build configuration.
README.md, lines 1-35	Install/run instructions, controls, known working features, and multiplayer/account limitations.
scripts/generate_characters.py	Helper script related to generating character assets; it is not the runtime world entry point.

Run the website

1. Install Node.js (npm comes with it).
2. Open a terminal in **C:/Users/S/Desktop/ChineseGame**.
3. Run **npm install** once to install dependencies.
4. Run **npm run dev** and open the local address printed by Vite.
5. Use the page navigation to enter Study Mode, Character Studio, or Social World.

What is connected today

The Study Mode, Character Studio, Social World, and profile share saved data through the profile module. The Social World is single-player. Real shared rooms, real online players, text chat, voice chat, moderation services, accounts, and cloud saves still require backend services and are not active in this project.